

COLD FORMED HOLLOW SECTIONS

STANDARD SPECIFICATIONS FOR WELDED CIRCULAR & NON-CIRCULAR STEEL TUBES

Standard Specification	Scope	Grade of Tubes	Chemical Composition Percentage (max. unless range is given)				Mechanical Properties			
			C	Mn	P	S	Condition	Tensile Strength (min.)	Yield Strength (min.)	Elongation (% min.)
1	2	3	4	5	6	7	8	9	10	11
ASTM-A53	E.R.W. Carbon Steel Pipes	Type E Grade A	0.25	0.95	0.05	0.06	As Rolled	ksi (MPa)	ksi (MPa)	(G.L. = 2" or 50mm)
								48 (331)	30 (207)	As per Table X7
ASTM-A120	E.R.W. Carbon Steel Pipes for ordinary use	Standard Pipe	--	--	--	--	--	--	--	--
ASTM-A135	Standard Specification for ERW Steel Pipe	Grade A	0.25	0.95	0.05	0.06	As Rolled	48 (331)	30 (207)	As per Table 1 Equation: (Elongation) = 56 x (Actual Thickness of Specimen, inch) + 17.50
ASTM-A500	E.R.W. Carbon Steel Structural Tubing in Round & Shapes	Round Grade A	0.26	--	0.04	0.05	As Rolled	45 (310)	33 (228)	25
		Grade B	0.26	--	0.04	0.05		58 (400)	42 (290)	23
		Shaped Grade A	0.26	--	0.04	0.05		45 (310)	39 (269)	25
		Grade B	0.26	--	0.04	0.05		58 (400)	46 (317)	23
ASTM-A513	E.R.W. Carbon Steel Mechanical Tubing	MT 1010	0.5-0.15	0.3-0.6	0.04	0.05	As Rolled	45 (310)	32 (221)	15
		MT 1015	0.1-0.2	0.3-0.6	0.04	0.05		48 (331)	35 (241)	15
		MT 1020	0.15-0.25	0.3-0.6	0.04	0.05		52 (359)	38 (262)	12
BS-31	E.R.W. Steel Conduits	Class A Plain End	--	--	--	--	As Rolled & Coated G.I. Strip	tonf/in ²	tonf/in ²	15 (G.L. = 8")
			--	--	--	--				
BS-980	E.R.W. Carbon Steel Pipes for Automobile Purposes	ERW 1	0.20	0.60	0.06	0.06	As Rolled	20	11	20
		ERW 2	0.30	0.60	0.06	0.06		25	15	15
		ERW 3	0.40	0.60	0.06	0.06		30	20	10 (G.L. = $4\sqrt{S_o}$)
BS-1139	E.R.W. Metal Scaffolding		--	--	0.06	0.06	As Rolled (Black Tubes)	20 - 30	13.5	$700/T.S.$ (ton/in ²) (G.L. = $4\sqrt{S_o}$)
BS-1387	E.R.W. Carbon Steel Tubes	Light	--	--	0.06	0.06	As Rolled	33 - 47	--	20 (G.L. = $5.65\sqrt{S_o}$)
		Medium	--	--	--	--				
		Heavy	--	--	--	--				
BS-1717	E.R.W. Carbon Steel Tubes for Bicycle and Motorcycle Purposes	ERW 101	0.10	0.60	0.06	0.06	As Rolled	20	11	--
		ERW 102	0.20	0.60	0.06	0.06		25	15	--
		ERW 103	0.30	0.60	0.06	0.06		30	20	--
BS-1775	E.R.W. Carbon Steel Tubes for Mechanical Structural and General Engineering Purposes	ERW 11	--	--	0.06	0.06	As Rolled	20	11	$700/T.S.$ (ton/in ²) (G.L. = $4\sqrt{S_o}$) or $600/T.S.$ (ton/in ²) (G.L. = $5.65\sqrt{S_o}$)
		ERW 16	--	--	0.06	0.06		25	16	
		ERW 20	--	--	0.06	0.06		30	20	
		ERW 23	--	--	0.06	0.06		32	23	

*1: G.L. = Gauge Length

*2: S_o = Original Cross-Section Area

*3: T.S. = Tensile Strength

*4: T.P. = Test Piece as per JIS Z2201

*5: H = Distance between Outside Surfaces

*6: D = Outside Diameter

*7: t = Wall Thickness of Tubes

*8: H' = Distance between Inside Surfaces